

reductions of 4.0 percent ($p = 0.030$) and salary declines of £476 ($p < 0.001$).

4.4 Robustness Checks

We conduct several robustness tests to assess the sensitivity of our results to alternative specifications and sample restrictions. First, we examine whether our findings depend on the fixed effects structure. Tables A7 and A8 present firm-level results using year-by-month fixed effects instead of industry-by-year-by-month fixed effects, while Table A9 replicates our occupation-level analysis using job category-by-period fixed effects. Despite these alternative specifications, all main conclusions persist: low-seniority employment declines at exposed firms, hiring decreases are, if anything larger, and occupation-level job postings and salaries fall significantly with exposure.

Second, we go one step further to address potential endogeneity concerns, and implement an extremely restrictive within-firm specification. To do so, we construct a firm-by-month-by-seniority panel that measures the number of people at low, medium, and high seniority levels working in a given firm in a given month.⁹ We then estimate a triple-difference model that interacts our post-ChatGPT dummy, firm-level LLM exposure, and seniority indicators, using the mid-seniority group as our reference group. This specification includes firm-by-period fixed effects, which absorb all time-varying firm-level shocks such as changes in profits, financing costs, industry conditions, management, or any other factor affecting the entire firm that might be correlated with both LLM exposure and employment decisions. We additionally include firm-by-seniority fixed effects (controlling for stable workforce composition preferences) and seniority-by-period fixed effects (capturing aggregate seniority-level trends). The results in Table A10 show strong evidence that employment declines concentrate in low-seniority positions: relative to mid-seniority workers, low-seniority employment at high-exposure firms contracts by an additional 0.7 percent ($p < 0.001$), while high-seniority positions show no significant differential effect. This result confirms the finding that LLM adoption disproportionately affects junior positions (see also Brynjolfsson, Chandar and Chen, 2025; Lichtinger and Hosseini Maasoum, 2025).

⁹Out of 7 seniority classifications in the Revelio Labs data, we classify the lowest two categories as junior (59% of jobs), the next two categories as mid (26%) and the highest three categories as senior (15%).

Third, we test whether our results depend on the minimum firm size threshold used in the sample construction of the firm-level data. Our main analysis restricts to firms with at least 100 employees to ensure consistent LinkedIn coverage and reliable estimation of exposure scores. Tables A11 to A13 relax this restriction to include all firms with at least 10 or 50 employees, expanding the sample to more than 11 million and 2.5 million firm-month observations. The qualitative patterns persist: the stock of low-seniority employment declines, average compensation increases, and exposed firms list significantly fewer jobs in both specifications, although the magnitudes are slightly different.

Fourth, we verify that our results are not sensitive to the task importance scheme used in our main specification. Recall that we follow Eloundou et al. (2024) by assigning weights of 1.0 to core tasks and 0.5 to supplemental tasks when calculating occupation-level exposure scores. Appendix A3.4 replicates our analyses using the continuous task importance scores provided in the O*NET database. The results are quantitatively and qualitatively similar to our main results, suggesting that our findings do not depend on which particular task weighting scheme we use.

Fifth, we estimate weighted regressions, weighing our observations in the firm-level regressions by pre-treatment employment, and in the occupation-level regressions by pre-treatment job posting volume. This approach gives greater weight to occupations representing larger shares of employment to make our estimates more representative for the experiences of the typical worker. Appendix A3.5 presents the results. The findings are largely in line with our main results. For firm-level employment stocks, the weighted specification shows similar overall employment effects, with a slightly larger decline in low-seniority employment (0.6 percent per standard deviation versus 0.4 percent unweighted). For firm-level job postings, the weighted coefficient remains negative but becomes statistically insignificant, suggesting smaller firms drive the hiring freeze response. For occupation-level outcomes, weighted estimates remain nearly identical: job postings decline by 3.8 percent (versus 3.9 percent unweighted) and posted salaries fall by £474 (versus £495 in the main specification). The overall consistency between weighted and unweighted specifications indicates our main conclusions are not driven by extreme outcomes at small entities, with the possible exception of our firm-level job listing result. Finally, we relax the assumption of linear treatment effects by categorizing occupations and firms into exposure quartiles. Tables A20 to A22 present results where the bottom

quartile serves as the reference category. For firm-level job postings, effects are significant across all higher quartiles, though the third and fourth quartiles show similar magnitudes, which is consistent with a threshold effect. For firm-level employment stocks, the evidence is weaker and non-monotonic, with only scattered significant effects across bins. The occupation-level analysis reveals clearer nonlinearities: while the second and third quartiles show minimal effects, the highest exposure quartile experiences significantly larger impacts, suggesting that most of the occupation-level results we observe are driven by the 25 percent most exposed occupations.

5 Conclusion

This paper provides evidence that large language models are reshaping UK labor markets. Using a difference-in-differences design that exploits variation in occupational- and firm-level LLM exposure, we conduct three analyses that show how labor markets change in response to the introduction of generative AI. First, we examine firm-level employment stocks to understand how workforce size and composition evolve within organizations. Second, we analyze firm-level hiring patterns to capture forward-looking recruitment decisions that precede employment adjustments. Third, we investigate occupation-level posting dynamics to identify economy-wide demand shifts.

Our firm-level employment analysis documents a 0.3 percent reduction in total employment per standard deviation increase in LLM exposure, though this effect only becomes statistically significant approximately 18 months after ChatGPT’s introduction. This reduction is predominantly concentrated in junior positions, which decline by 0.4 percent per standard deviation, while senior positions show a smaller, statistically insignificant 0.2 percent reduction. The resulting compositional shift reduces the fraction of junior workers by 0.1 percentage point per standard deviation, mechanically increasing average compensation by £106 as firms retain proportionally more higher-paid senior staff, though this compensation effect is only marginally significant.

While employment stocks adjust gradually, job posting data provide real-time evidence of firms’ evolving labor demand. Firm-level posting activity drops following

ChatGPT’s release, with a one standard deviation increase in LLM exposure reducing the probability of posting any vacancy by 1.0 percentage point. This reduction emerges gradually and becomes significant after approximately 10 months, mirroring the timeline of employment stock adjustments. When we examine the composition of roles driving this reduction, we find that technical positions and creative roles experience the sharpest declines. Sales representative positions, by contrast, show a statistically significant increase, suggesting that customer-facing roles requiring direct interpersonal interaction remain robust to LLM-based automation.

In addition to within-firm dynamics, our occupation-level analysis reveals how labor demand shifts for entire job categories regardless of firm exposure. A one standard deviation increase in LLM exposure is associated with a 3.9 percent reduction in posting volumes, accompanied by £495 (1.1 percent) salary declines. The apparent inconsistency of declining posted salaries for exposed occupations alongside increasing firm-level compensation shows how aggregation could mask opposing forces. While LLM exposure drives down market wages for specific occupations, within-firm rebalancing toward more senior positions pushes average firm-level wages in the opposite direction.

Our heterogeneity analysis reveals that the observed labor market disruptions are almost entirely driven by high-remuneration segments of the labor market. This concentration of effects in higher-wage segments contradicts historical automation patterns that predominantly displaced routine middle-skill work, and suggests that LLMs might lead to a compression of the wage distribution.

Our findings carry important implications for understanding technological displacement and designing policy responses. The concentration of job losses among entry-level positions disrupts traditional skill development pathways where workers master progressively complex tasks through hands-on experience. Without junior roles serving as training grounds, firms may struggle to develop senior talent internally while new entrants face shrinking opportunities to gain experience (Ide, 2025). The freeze in hiring followed by employment stock changes suggests that forward-looking indicators like job postings may provide early warning signals for policymakers. The sharp reductions in technical and analytical roles alongside resilience in sales positions indicates that human interaction and relationship-building skills remain valuable complements to LLM capabilities, suggesting that retraining programs should emphasize these interpersonal competencies. Policy responses